

Add filtering for chat template kwargs

#25794

New issue

Merged

DarkLight1337 merged 7 commits into `vllm-project:main` from `russellb:vllm-ghsa-6fvq-23cw-5628` on Sep 27, 2025

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Files changed5

+158-6

russellb commented on Sep 27, 2025Member

• edited by github-actionsbot

- [GHSA-6fvq-23cw-5628](#)
- <https://github.com/vllm-project/vllm/security/advisories/GHSA-4hw3-g54f-xfrj>

Isotr0py and others added 4 commits 4 months ago

[untrust chat template from request by default](#) [cf29ddd](#)

[avoid chat template kwargs attack](#) [4936dbe](#)

[add chat template kwargs test](#) [04c5021](#)

Merge remote-tracking branch 'origin/main' into vllm-[GHSA-6fvq-23cw-5628](#) [30952a9](#)

russellb requested review from **DarkLight1337**, **NickLucche**, **aarnphm**, **chaunceyjiang**, **robertgshaw2-redhat** and **simon-mo** as [code owners](#) 4 months ago

russellb added this to the **v0.11.0 Cherry Picks** milestone on Sep 27, 2025

mergifybot added the **frontend** label on Sep 27, 2025

Reviewers

youkaichao

Isotr0py

DarkLight1337

robertgshaw2-redhat

simon-mo

aarnphm

NickLucche

chaunceyjiang

+3 more reviewers

wuwenbin

Eviannn

gemini-code-assist[bot]

Assignees

No one assigned

Labels

frontend

ready

Projects

None yet

Milestone

v0.11.0 Cherry Picks



gemini-code-assist (bot) reviewed [on Sep 27, 2025](#)

[View reviewed changes](#)

Development

Successfully merging this pull request may close these issues.

None yet

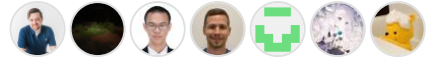
gemini-code-assist (bot) left a comment

Contributor

Code Review

This pull request addresses a critical security vulnerability ([GHSA-6fvq-23cw-5628](#)) related to arbitrary code execution via malicious chat templates. The core of the fix is the new `resolve_chat_template_kwargs` function, which intelligently filters keyword arguments passed to the Jinja2 template renderer. It ensures that only arguments both supported by the `apply_chat_template` function and explicitly declared as variables within the template are passed, effectively blocking the injection of a malicious `chat_template`. The changes also introduce a `--trust-request-chat_template` flag for defense-in-depth, preventing user-supplied templates from being used unless explicitly allowed. The implementation is robust, and the accompanying tests correctly validate the fix. Overall, this is a solid and important security patch.

7 participants



DarkLight1337 approved these changes [on Sep 27, 2025](#)

[View reviewed changes](#)



DarkLight1337 added the `ready` label [on Sep 27, 2025](#)



DarkLight1337 commented [on Sep 27, 2025](#)

Member

[@Isotr0py](#) PTAL at the V1 test failure



DarkLight1337 commented [on Sep 27, 2025](#)

Member

Let's also reject the request if it passes an untrusted chat template, to avoid silent regressions



Isotr0py added 3 commits [4 months ago](#)



[refuse request with untrust chat template](#) ...

[70f1d84](#)



[fix api server test](#) ...

[95405ad](#)



[fix v1 entrypoint^Cest](#) ...

✓ [a641e9a](#)



DarkLight1337 enabled auto-merge (squash)
[4 months ago](#)



DarkLight1337 merged commit **7977e50**
into [vllm-project:main](#) on [Sep 27, 2025](#)
44 checks passed

[View details](#)



simon-mo pushed a commit that referenced this pull request
on [Sep 28, 2025](#)



[Add filtering for chat template](#)
[kwargs \(#25794\)](#) ...

[04c2b26](#)



pdasigi pushed a commit to [pdasigi/vllm](#) that referenced this
pull request on [Oct 3, 2025](#)



[Add filtering for chat template](#)
[kwargs \(vllm-project#25794\)](#) ...

[04df893](#)



yewentao256 pushed a commit that referenced this pull
request on [Oct 4, 2025](#)



[Add filtering for chat template](#)
[kwargs \(#25794\)](#) ...

[1cb6005](#)



youkaichao reviewed
on [Oct 4, 2025](#)

[View reviewed changes](#)

[vllm/entrypoints/chat_utils.py](#)

Comment on lines [+1621](#) to [+1625](#)

```
1621 + resolved_kwargs = resolve_cl
1622 + tokenizer=tokenizer,
1623 + chat_template=hf_chat_te
1624 + chat_template_kwargs=kw
1625 + )
```



youkaichao on [Oct 4, 2025](#)

[Member](#)

is it called by each request? do we need to cache
it?



Isotr0py on [Oct 4, 2025](#)

[Member](#)

Yea, it's called by each request. This function
won't compile the chat template, so I think it won't
introduces too much overhead.

Given that `kwargs` can be various depending on the request's extra body, it's not really applicable to cache it.



Isotr0py on [Oct 4, 2025](#)

Member

Add cache in [#26227](#)



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Eviannn on [Dec 2, 2025](#)

Contributor

Yea, it's called by each request. This function won't compile the chat template, so I think it won't introduces too much overhead.

Given that `kwargs` can be various depending on the request's extra body, it's not really applicable to cache it.

using npu(ascend 910c) it costs about 10-20ms, it is great to cache it



wuwenbin on [Dec 23, 2025](#)

It seems this change might be linked to a ~10x latency jump we saw in our benchmarks on 0.11.0 (where CPU hit 100%).

The caching in 0.11.1 appears to solve it, but upgrading is a bit tricky since it requires CUDA $\geq$ 12.9.



DarkLight1337 on [Dec 23, 2025](#)

Member

You can still use older CUDA versions if you install vLLM from source.



1



youkaichao commented on [Oct 4, 2025](#)

Member

in which case do we ever use chat template from users? I thought we should just disallow it.



bbrowning commented on [Oct 4, 2025](#)

Contributor

We may also want a similar `trust_request_chat_template` around

[vllm/vllm/entrypoints/openai/serving_tokenization.py](#)

Line 83 in [2a6dc67](#)

83 `chat_template=request.chat_template or self`



Isotr0py mentioned this pull request [on Oct 4, 2025](#)

[Frontend] Cache chat template

Merged

kwargs resolution #26227

5 tasks



xuebwang-amd pushed a commit to xuebwang-amd/vllm that referenced this pull request [on Oct 10, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[2c589ca](#)



choprahetarth pushed a commit to Tandemn-Labs/vllm that referenced this pull request [on Oct 12, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[f789bda](#)



shyeh25 pushed a commit to shyeh25/vllm that referenced this pull request [on Oct 14, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[9062fb1](#)



lywa1998 pushed a commit to lywa1998/vllm that referenced this pull request [on Oct 20, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[8c3930f](#)



alhrido pushed a commit to alhrido/vllm that referenced this pull request [on Oct 24, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[7637bba](#)



xuebwang-amd pushed a commit to xuebwang-amd/vllm that referenced this pull request [on Oct 24, 2025](#)



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

[1e2d10b](#)



wangln19 mentioned this pull request [on Oct 28, 2025](#)

fix: allow HuggingFace standard chat
template params via **kwargs #27622

Merged

5 tasks



rtourgeman pushed a commit to rtourgeman/vllm that
referenced this pull request on Nov 10, 2025



Add filtering for chat template
kwargs ([vllm-project#25794](#)) ...

971d41a